CAPTURING THE DESENSITIZED CONFORMATION OF THE MUSCLE-TYPE NICOTINIC ACETYLCHOLINE RECEPTOR

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ABSTRACT

The nicotinic acetylcholine receptor (nAChR) from electric fish is a key model for understanding ligand-gated ion channel (pLGIC) function. Upon neurotransmitter binding, these receptors undergo successive conformational transitions from resting (closed-channel) to active (open-channel), and then desensitized (closed-channel) states. Recent structural data have clarified the coupling between agonist-induced changes in the extracellular domain (ECD) and motions of the pore-lining M2 α-helices in the transmembrane domain (TMD)[1,2]. However, agonist-bound *Torpedo* nAChR structures revealed lipid densities in the central pore, likely artefacts from reconstitution in protein-based nanodiscs[3]. Moreover, molecular dynamics simulations show that these structures adopt a hydrated, ion-permeable pore, inconsistent with desensitization.

To address this, we used amphipathic polymers (CyclAPol C_8-C_0-50)[4,5] to directly extract nAChR from fish electric organ membranes. This allows us to bypass reconstitution steps and preserve the native lipid context, which proves to be enriched in docosahexaenoic acid (DHA)-containing phospholipids. The cryo-EM structure of amphipol-purified, agonist-bound nAChR differs from those obtained in protein-based nanodiscs: while agonist-bound ECD signatures are conserved, the TMD shows an α-M2 helix break, leading to the closure of hydrophobic pore gates, consistent with the expected desensitized state. This highlights the impact of the amphipathic environment on membrane protein structural studies and the need to

compare different solubilization strategies to assess whether a given conformation is mimetic-specific or robust.

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